Configuring API-KEY

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5. Configuring the multi_brains Feature Package API
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1. Course Content

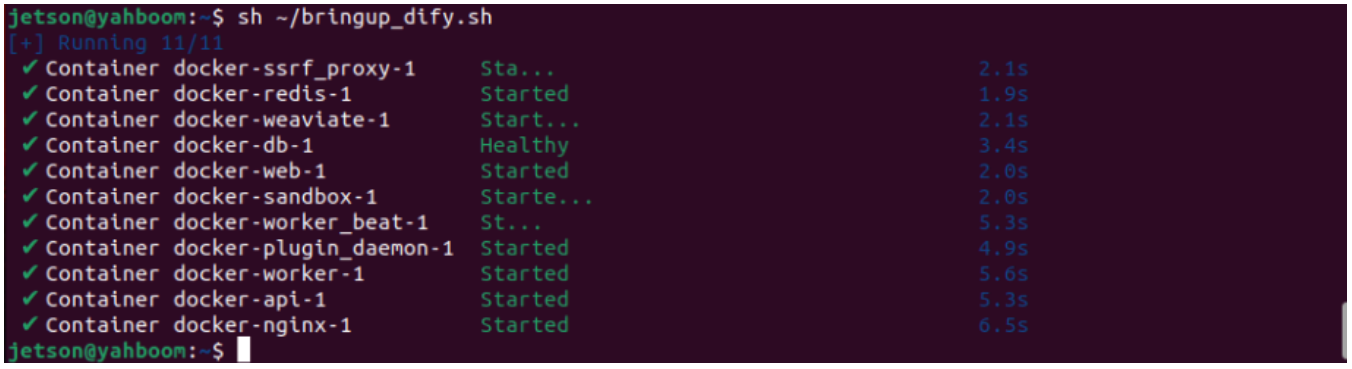
- Use the API-KEY registered earlier to configure the API-KEY on the robot's onboard computer.

> ⚠ **Warning**
>
> **Note**: Make sure the onboard computer is connected to the Internet so that the cloud model services can be used.

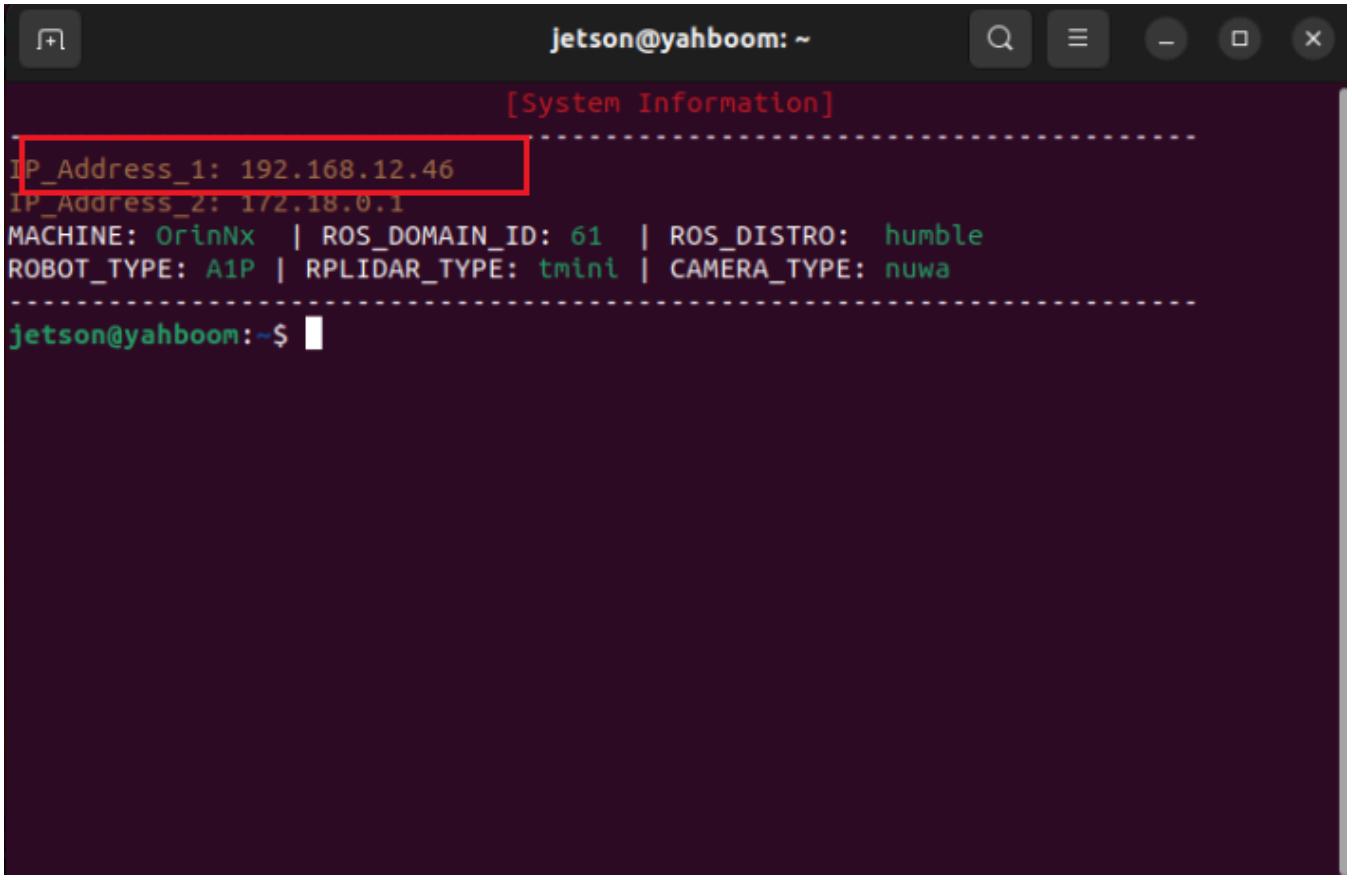
2. Starting the Dify Service

- Connect to the onboard computer via VNC or SSH, and enter the following command in the terminal:

```
sh ~/bringup_dify.sh
```



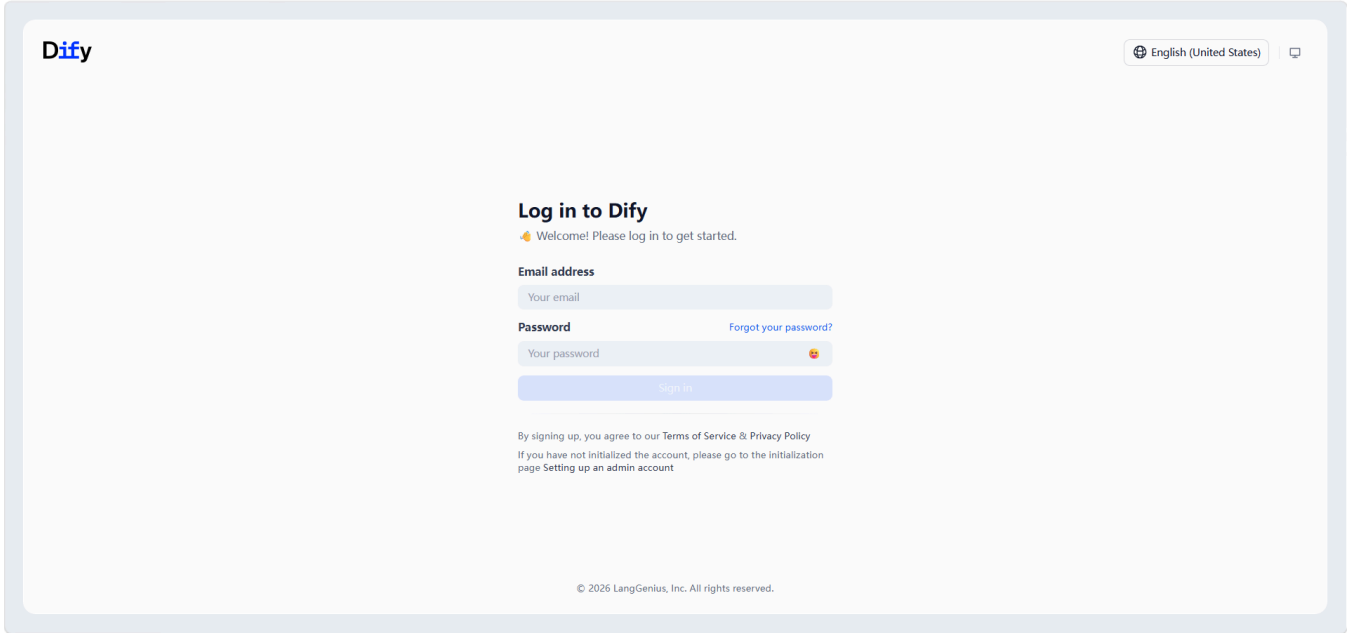
- Open a new terminal and you can see the IP address of this machine.

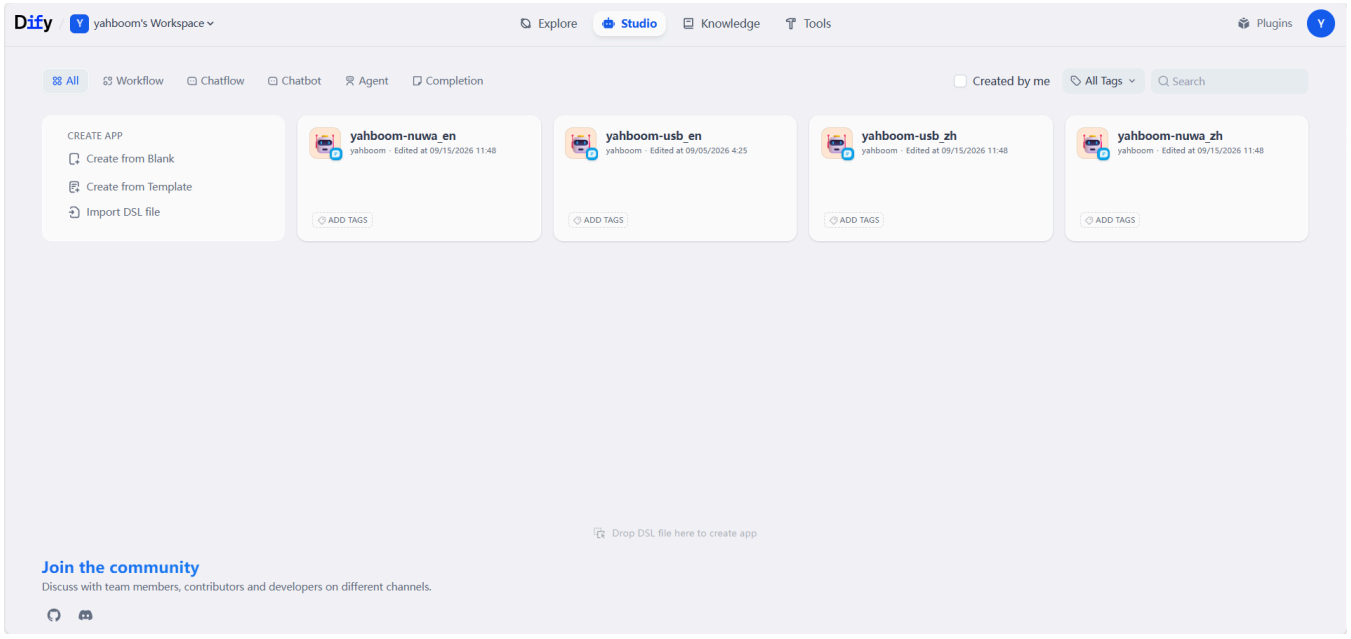


- Enter the IP address of the onboard computer directly in the browser address bar to access the Dify management page. If the device is being logged in to for the first time, you will need to use the account and password. You can select the language in the top-left corner.

> **Note**
>
> - Account name: `yahboom@163.com`
> - Password: `yahboom123`
> - All account names, passwords, agent applications, and RAG data are stored locally.

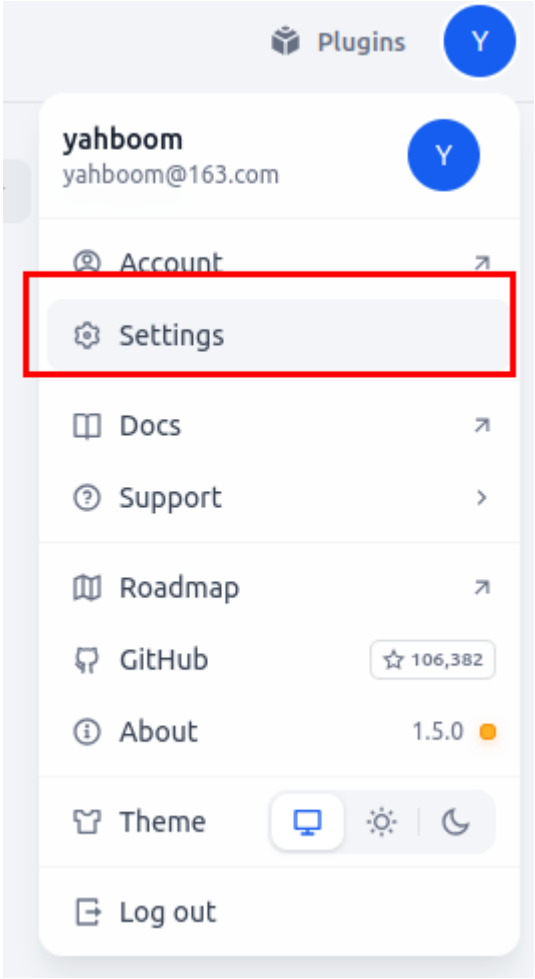
- After logging in, the page is as shown below.



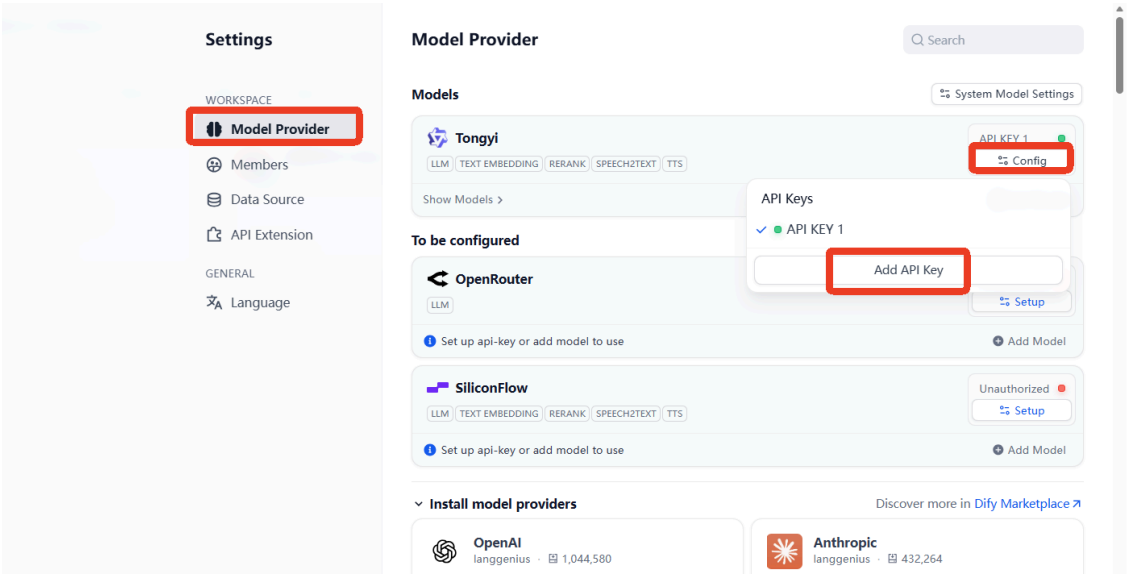


3. Configuring the Model Provider API-KEY

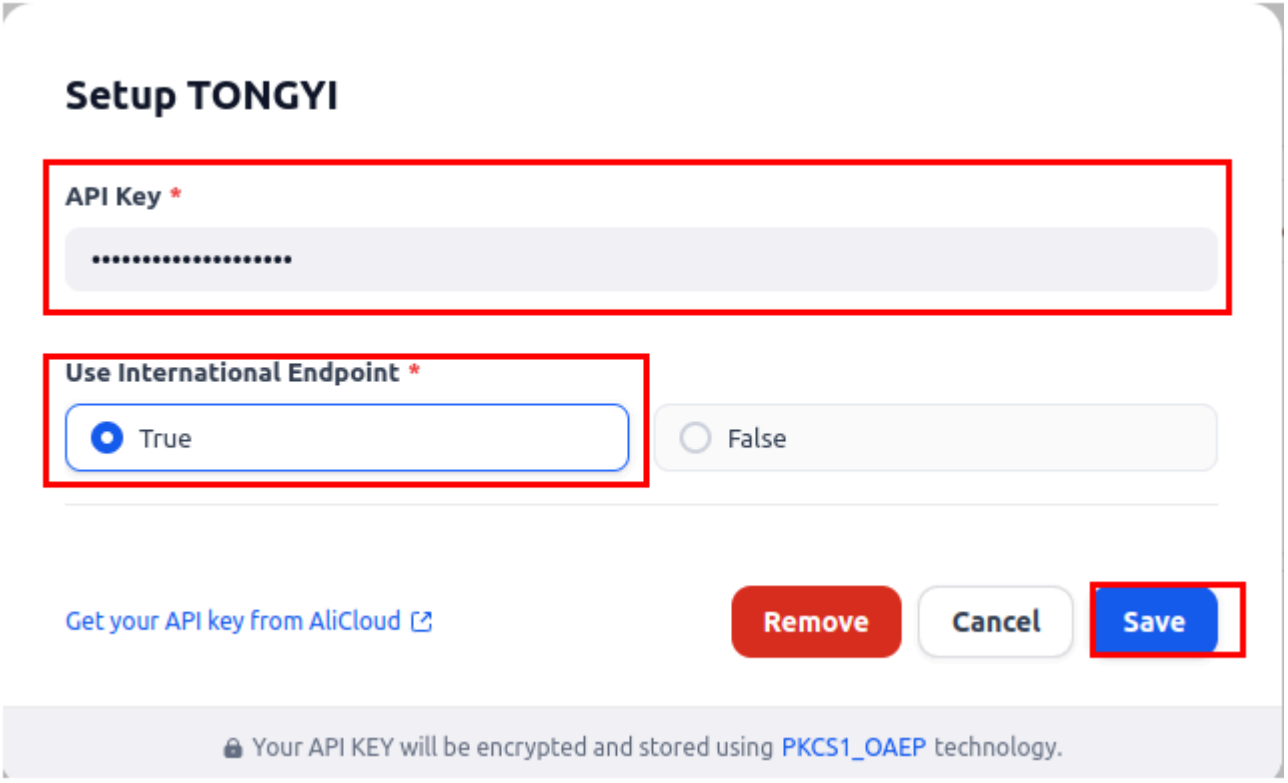
- Click Settings.



- Here, configuring an API for a Bailian account is used as an example. Click Model Provider → Config → Add API Key.



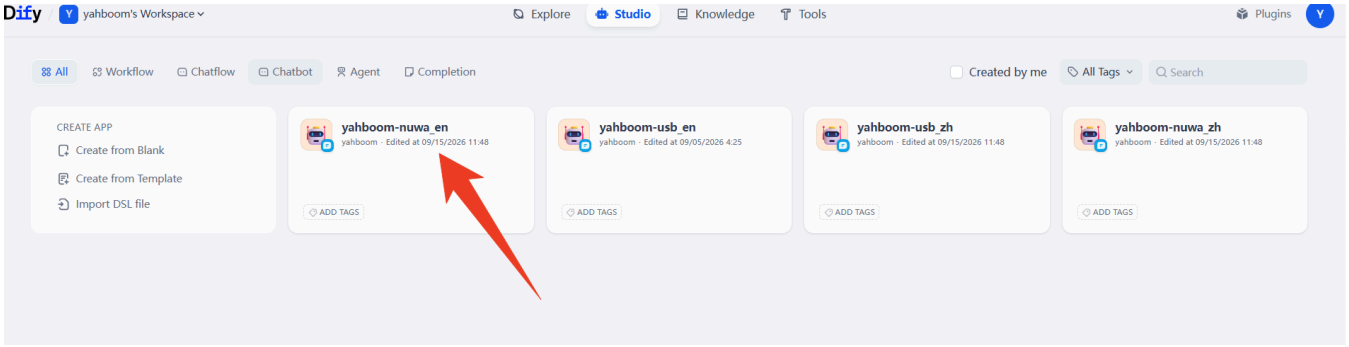
- Fill in the API-KEY from your Bailian platform account, then select whether it is an international site account, and click Save.



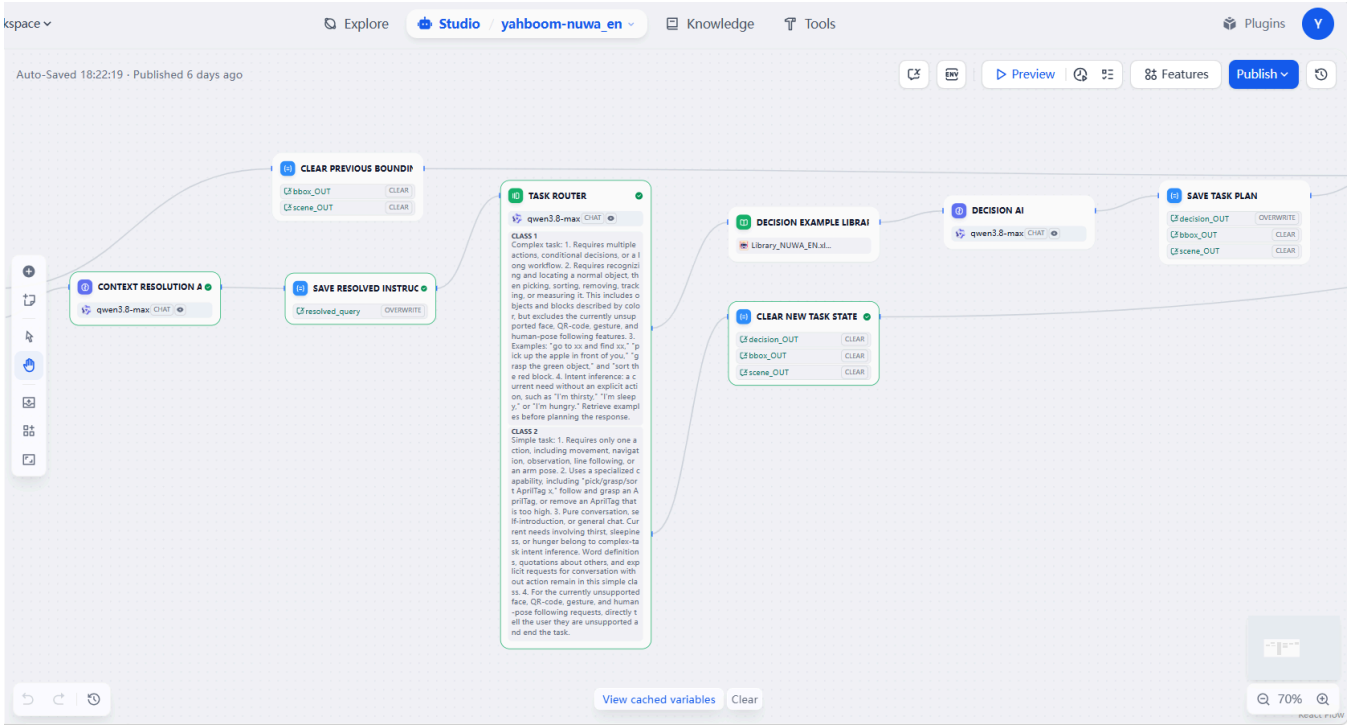
4. Testing the API-KEY

> **Tip**
>
> - If you need to test whether the API-KEY is available, you can refer to the tutorial in this section. If not, you can ignore it.

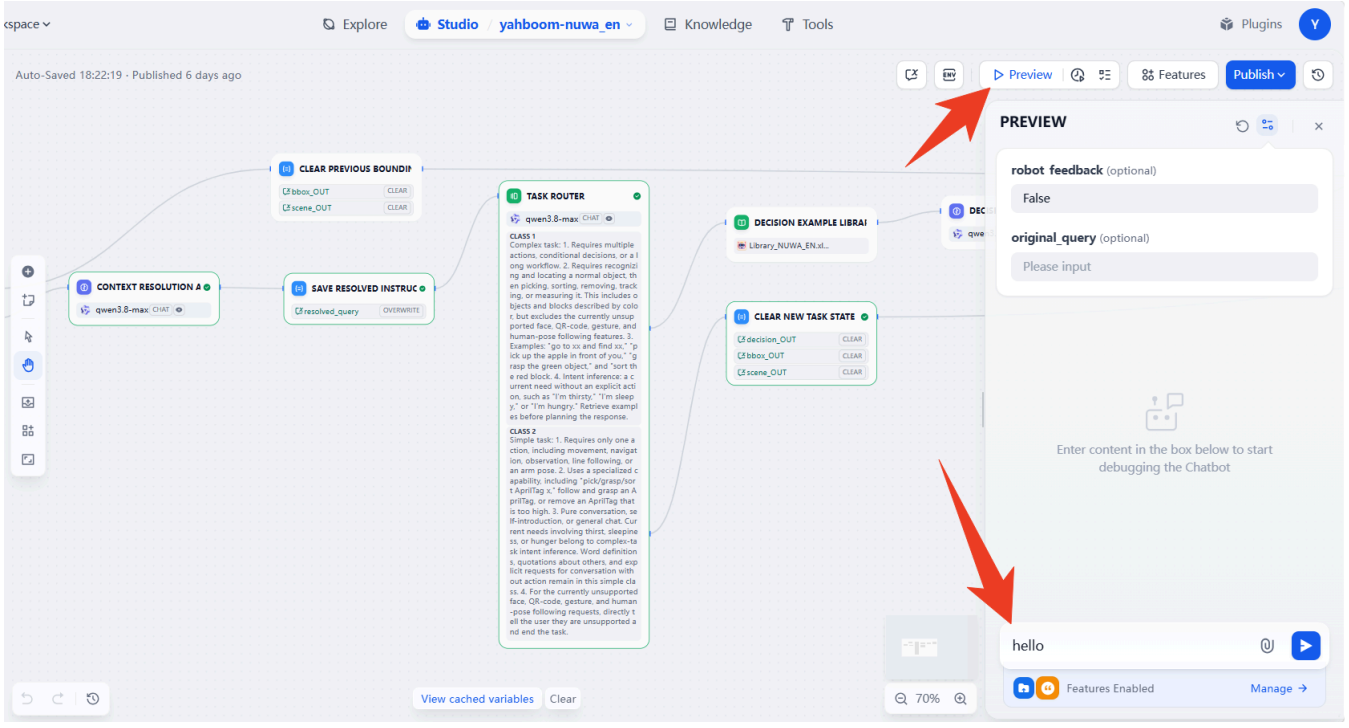
- Click the `yahboom-nuwa_en` application in the Studio.



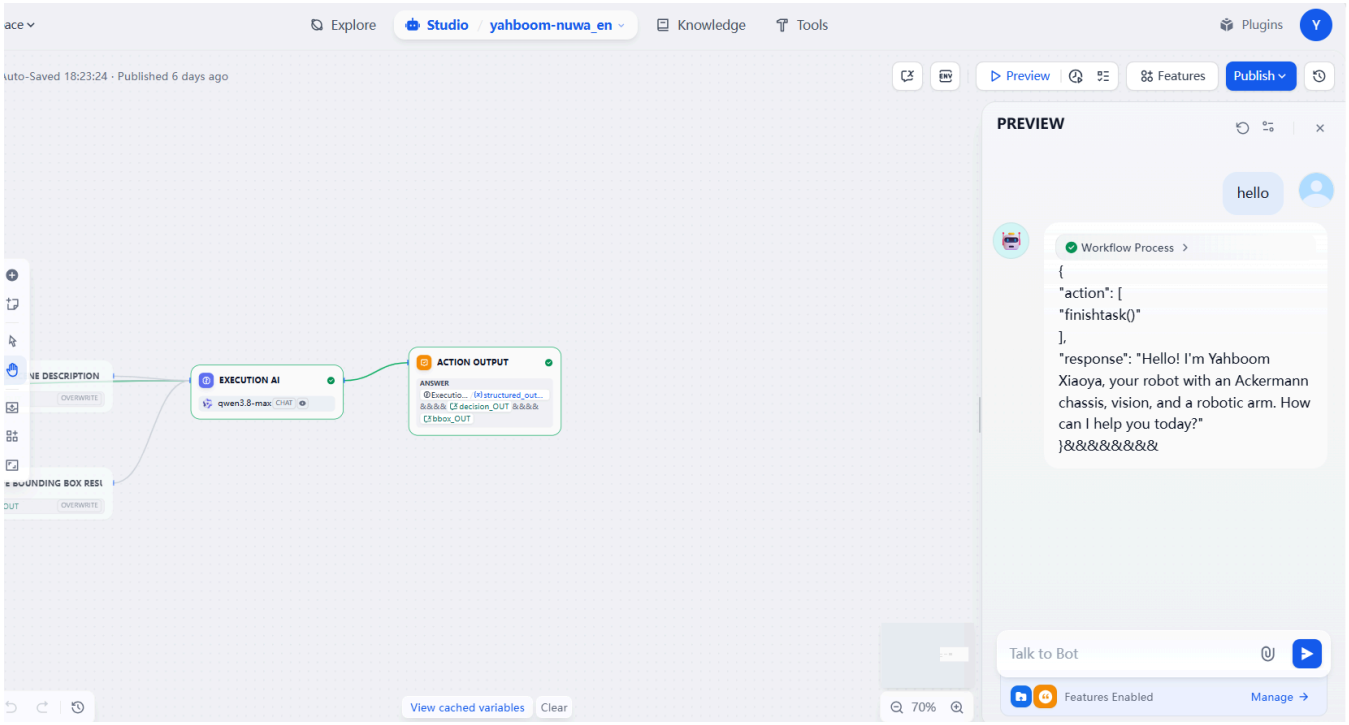
- After entering, you can see the workflow screen.



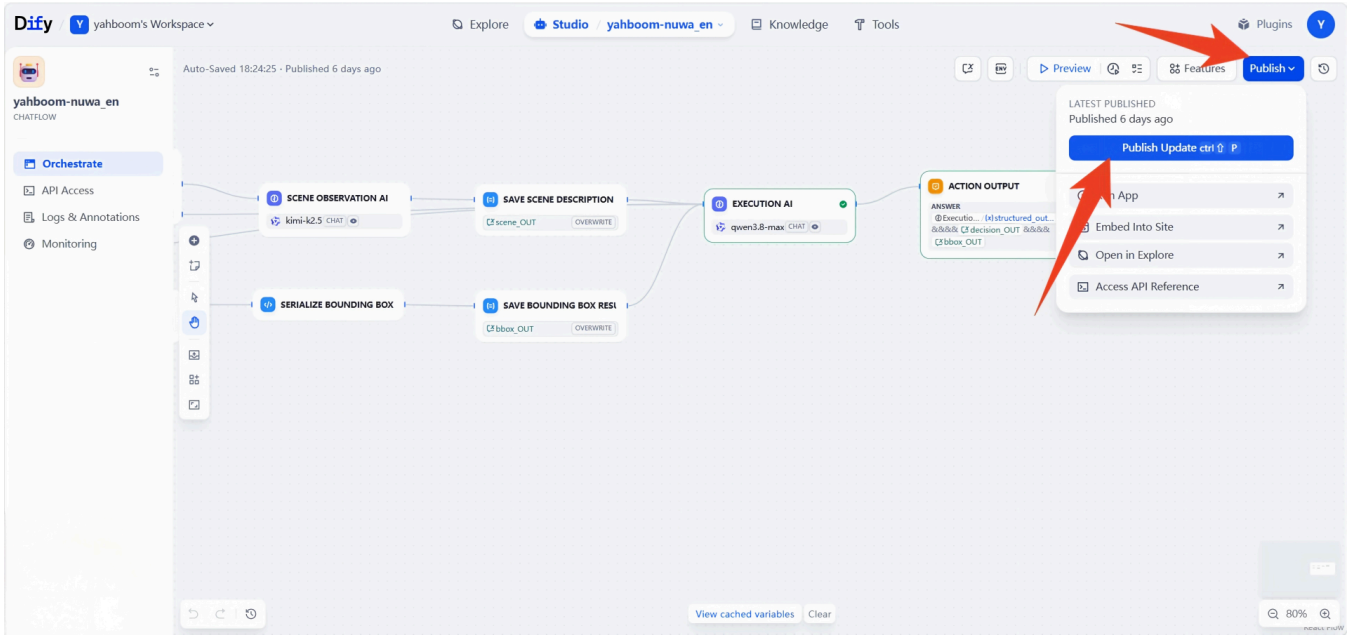
- Click Preview, then enter `Hello` and press Enter.



You can see the agent's reply.

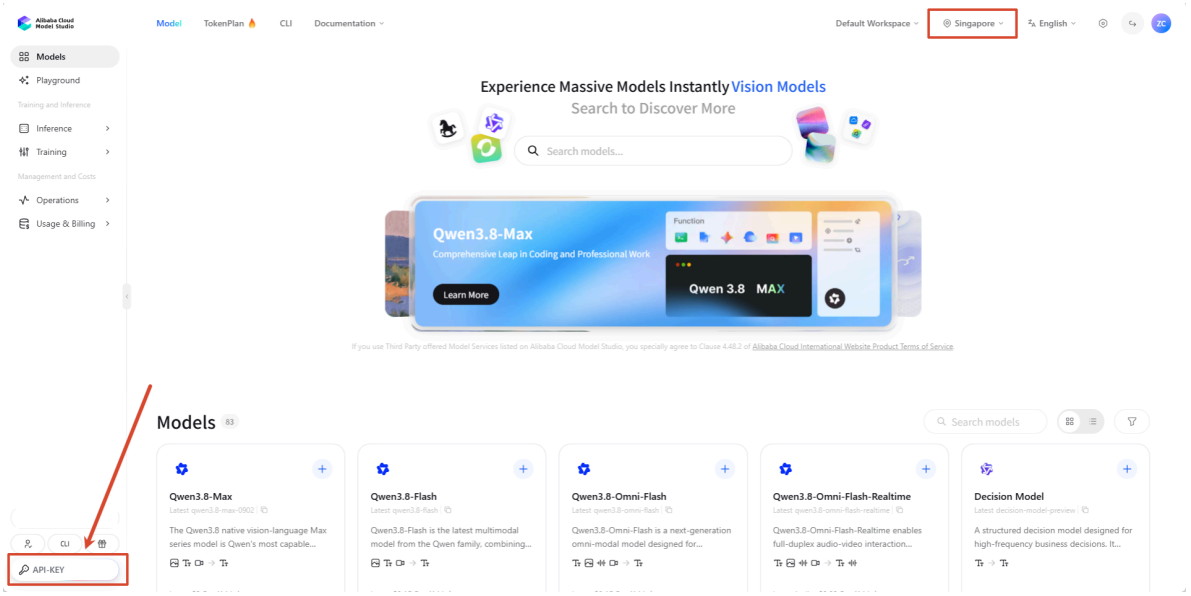


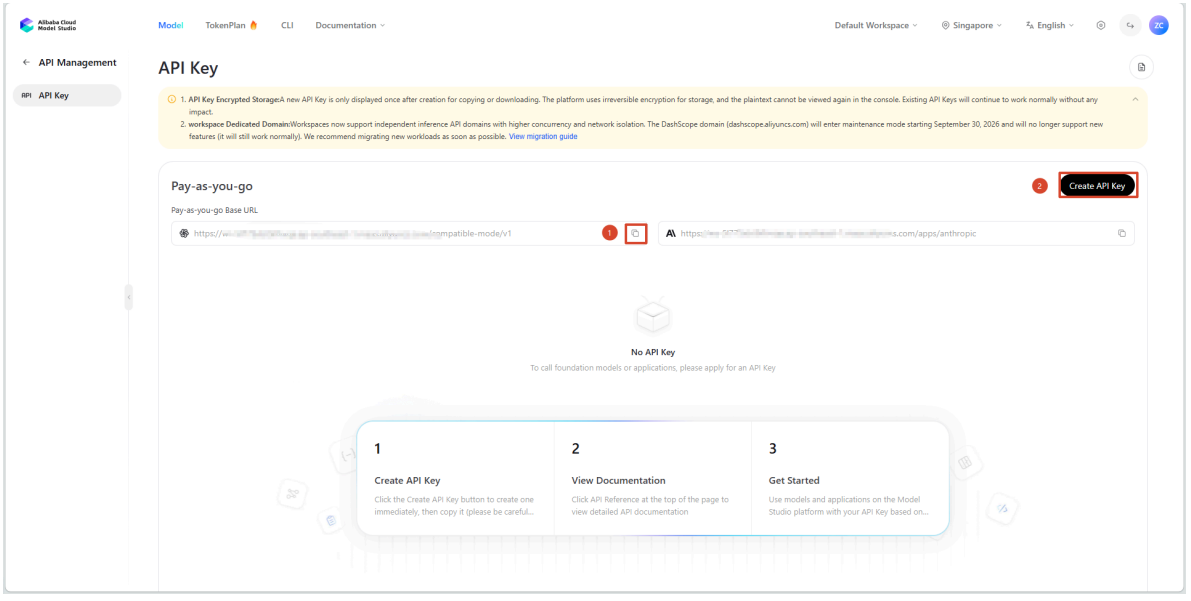
- If you have modified the model or the description inside the agent, you need to click Publish again.

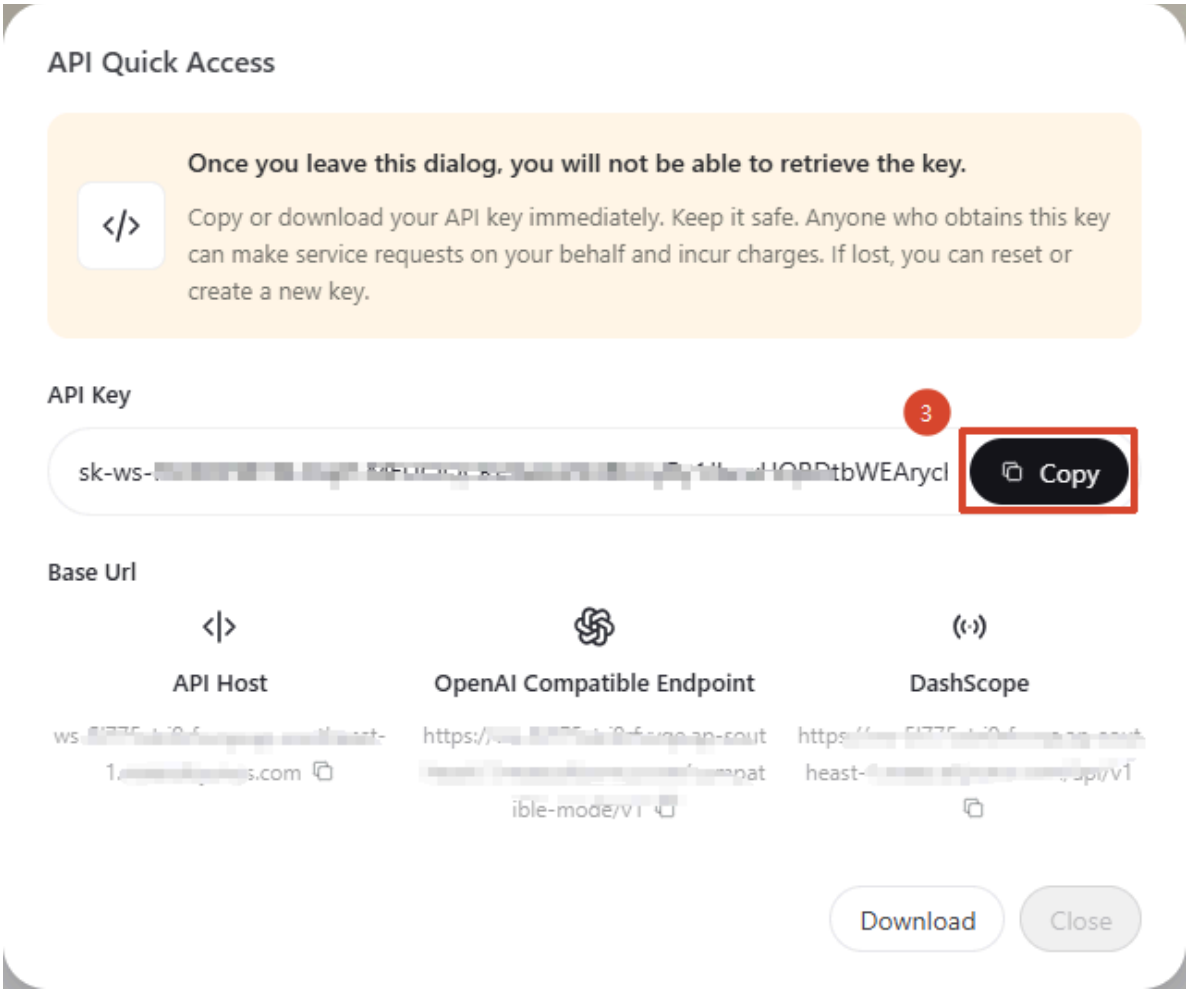


5. Configuring the multi_brains Feature Package API

- Fill in the Base URL and API Key (refer to the accompanying tutorial【AI Model Development】-【1.Register a model service provider account】).
    - Users need to configure the Base URL corresponding to their own account. Go to the Alibaba Cloud Model Studio Console (China site: Alibaba Cloud Bailian Console), apply for an API Key, copy the API Key and the OpenAI-compatible address (Base URL), and fill them into the corresponding fields.

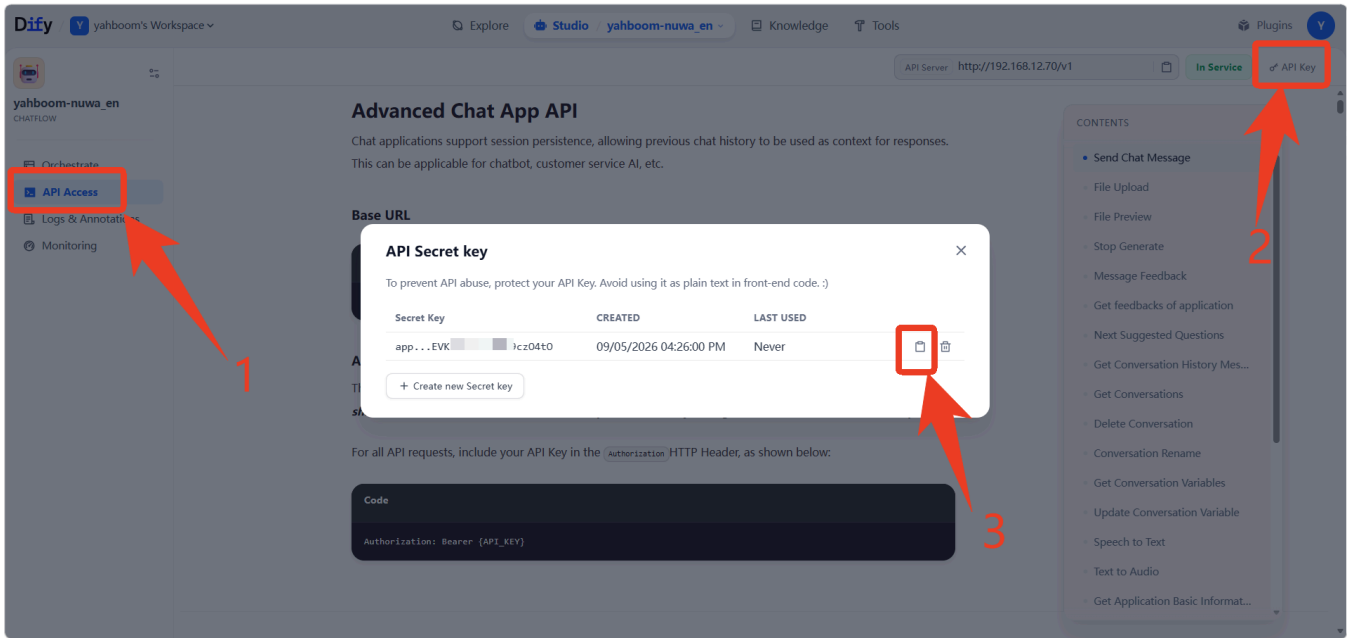

Model
TokenPlan
CLI
Documentation
Default Workspace
Singapore
English
Models
Playground
Inference
Training
Operations
Usage & Billing
Experience Massive Models Instantly Vision Models
Search to Discover More
Search models...
Qwen3.8-Max
Comprehensive Leap in Coding and Professional Work
Learn More
Function
Qwen 3.8 MAX
Models
Search models
Qwen3.8-Max
Qwen3.8-Flash
Qwen3.8-Omni-Flash
Qwen3.8-Omni-Flash-Realtime
Decision Model
CLI
API-KEY


Model
TokenPlan
CLI
Documentation
Default Workspace
Singapore
English
API Management
API Key
API Key
Pay-as-you-go
Pay-as-you-go Base URL
Create API Key
No API Key
To call foundation models or applications, please apply for an API Key
1
Create API Key
2
View Documentation
3
Get Started



> **Important**
>
> The key information of the Dify application `yahboom-nuwa_en` is configured by default at the factory, so there is no need to configure it again. If you want to switch to another agent, refer to the content below.

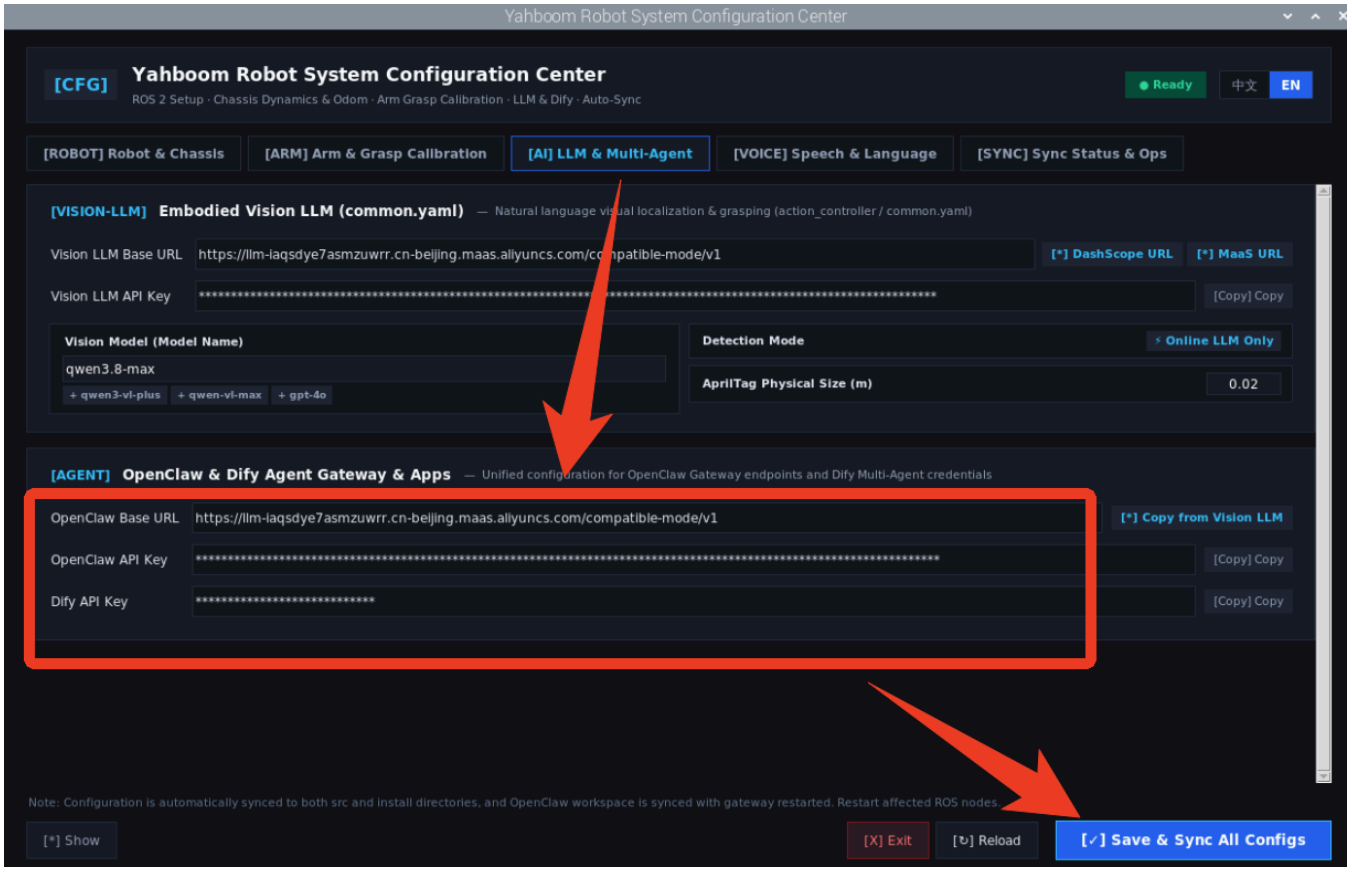
- You can find the current Dify Key in the Dify backend. `DIFY_API_KEY` is the API access credential used by the `multi_brains` program to access the Dify application.



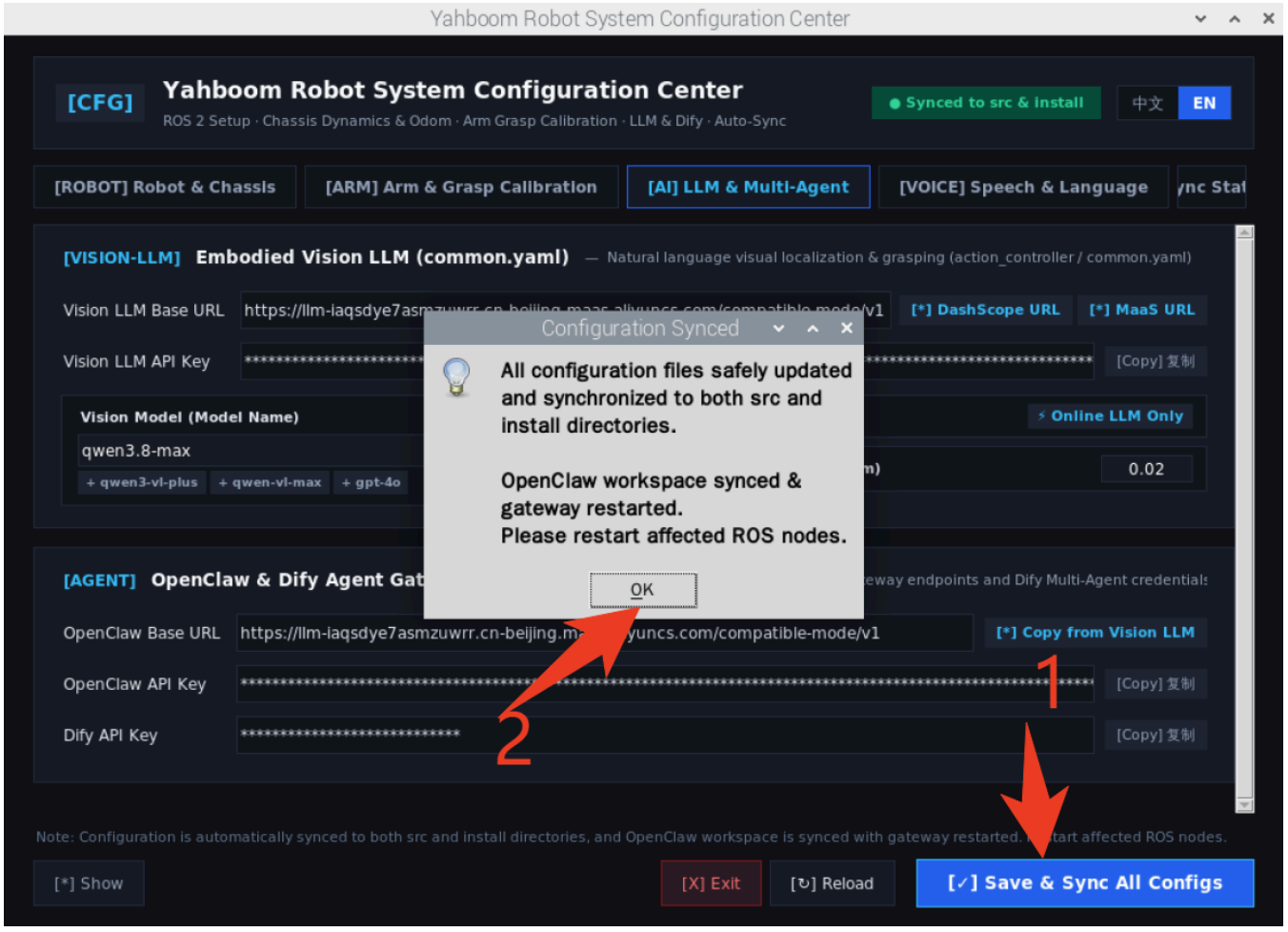
- Open the `app` application on the desktop, find Dify Agent Application — `DIFY_API_KEY`, and paste the API Key copied in the previous step into it.



You can also fill in the Base URL and API Key you just applied for. These are used for online speech recognition and speech synthesis.

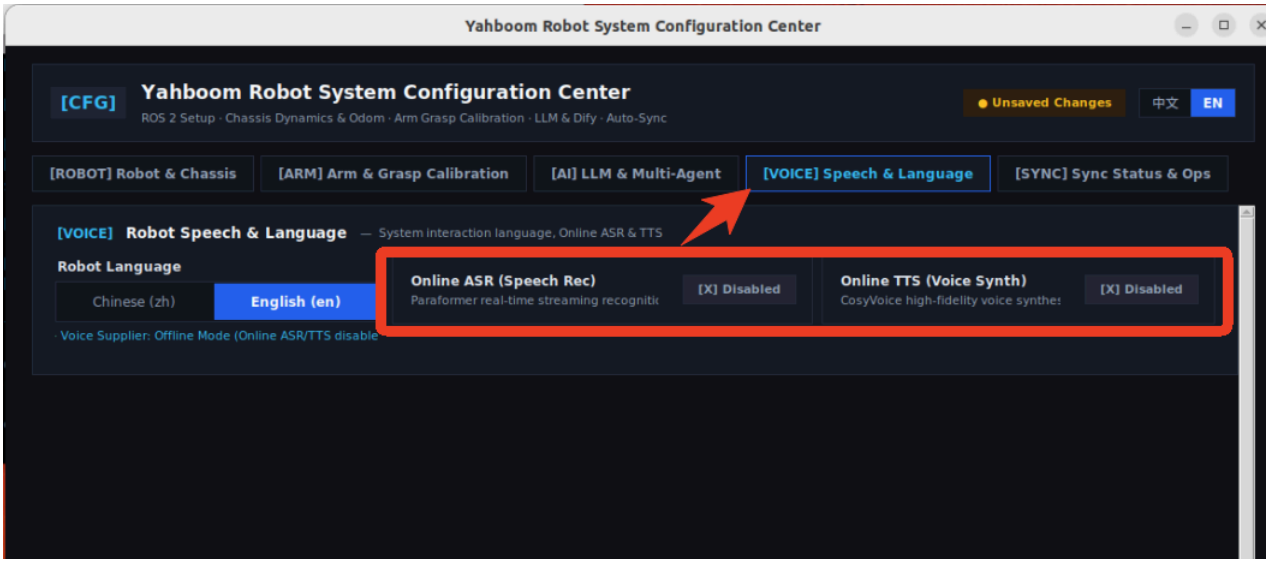


After clicking Save & Sync All Configs, you can exit.

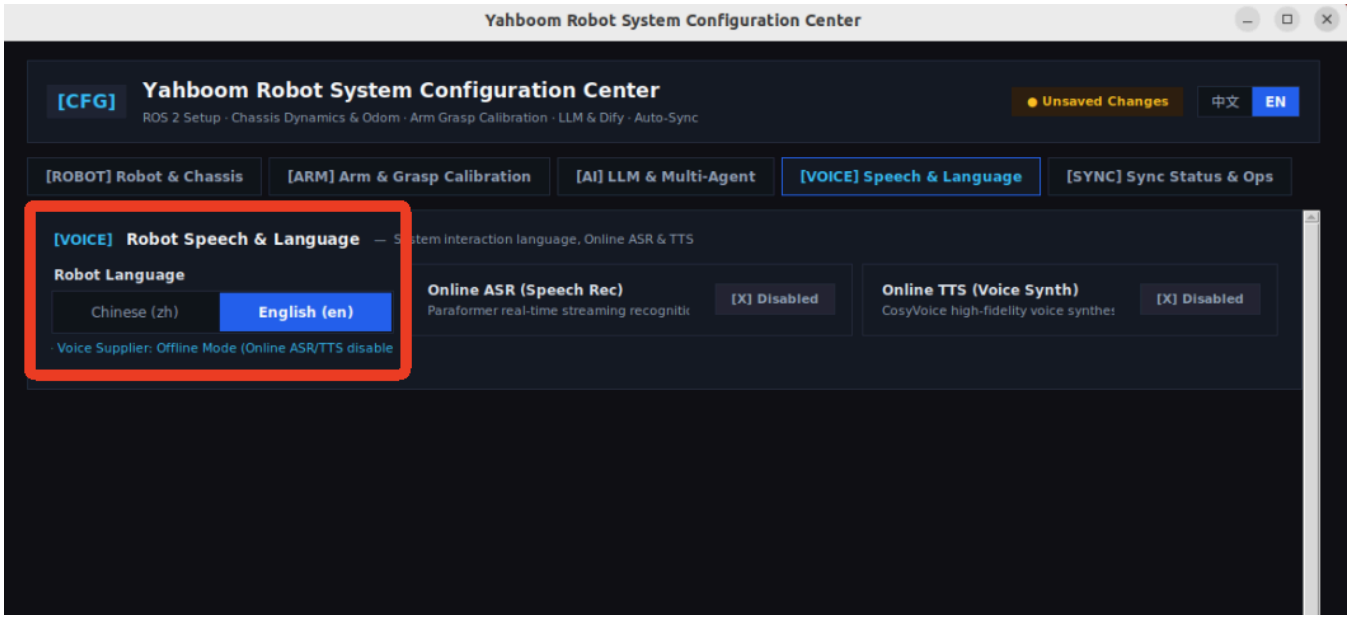


Modifying Other Parameters

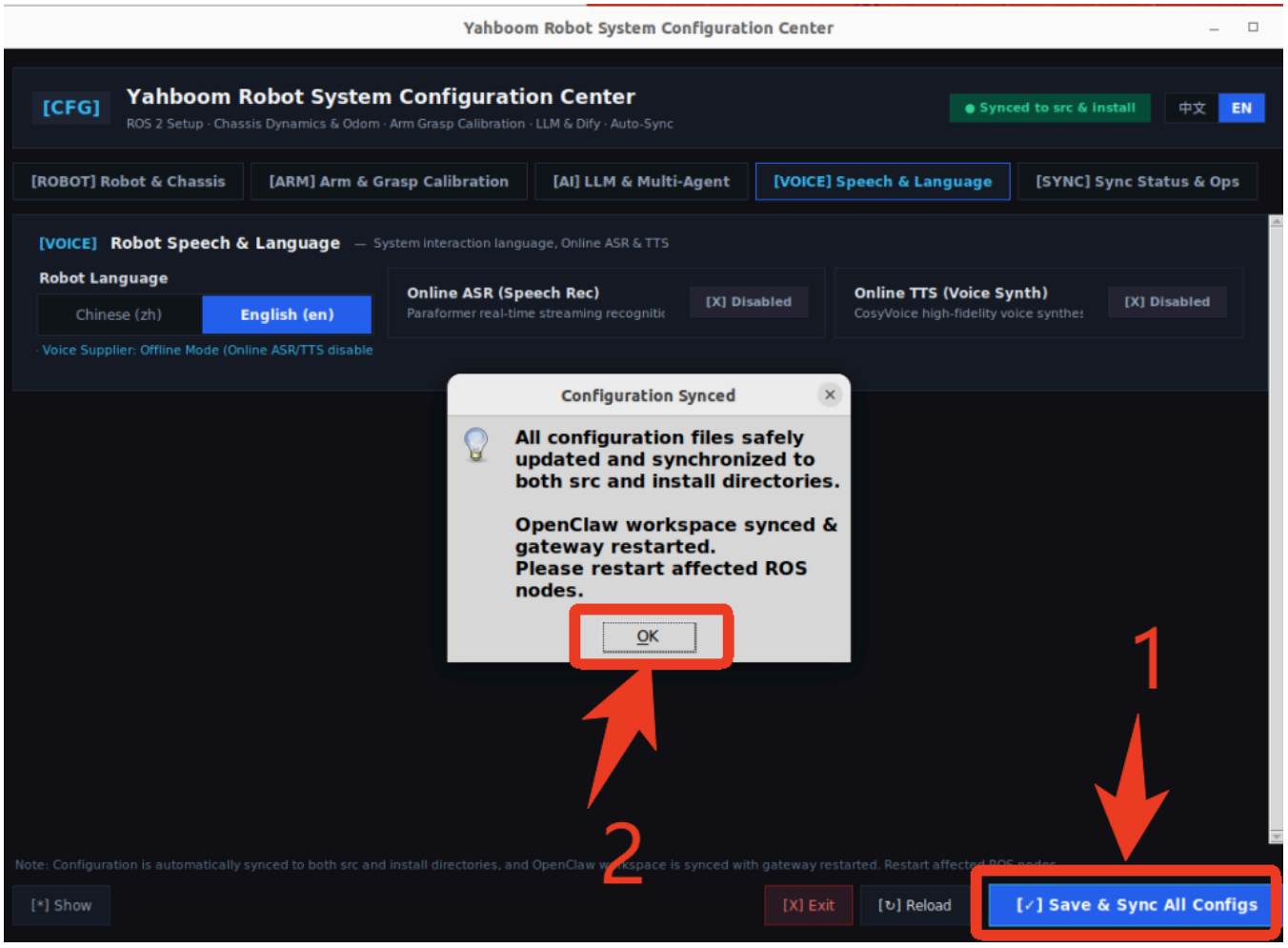
- If you need to use local speech recognition and speech synthesis, you can choose to disable the online services in the `app`.
    - Once online speech recognition and synthesis are disabled, the local services can be used.



- You can also choose domestic (China) or international (English).
    - For an international language selection, use the international (English) speech recognition and speech synthesis.



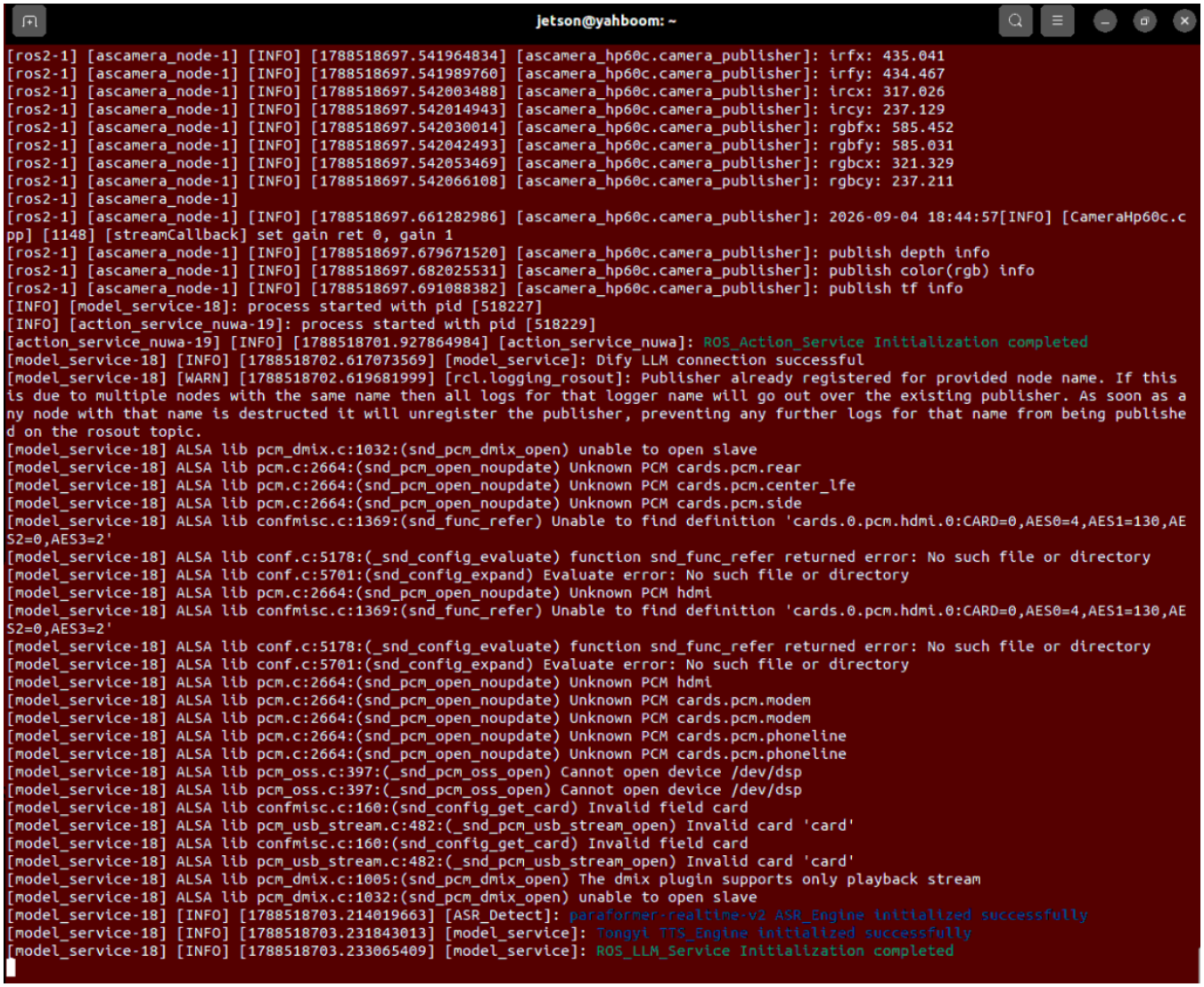
- After making the changes, save & Sync All Configs, and then exit.



6. Starting Voice Interaction

- Start the agent core node:

```
ros2 launch multi_brains llm_agent_control.launch.py
```



- Use the wake-up command `Hi,yahboom` to start interaction.

```
[model_service-18] [INFO] [1788518703.231843013] [model_service]: Tongyi TTS_Engine initialized successfully
[model_service-18] [INFO] [1788518703.233065409] [model_service]: ROS_LLM_Service Initialization completed
[model_service-18] [INFO] [1788518759.103486133] [ASR_Detect]: I'm here
[action_service_nuwa-19] [INFO] [1788518759.108356646] [action_service_nuwa]: wakeup received
[model_service-18] [INFO] [1788518761.961600954] [ASR_Detect]: 1-1-1
[model_service-18] [INFO] [1788518761.963627145] [ASR_Detect]: 1-1-1
[model_service-18] [INFO] [1788518761.965355625] [ASR_Detect]: 1-1-1
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[model_service-18] [INFO] [1788518761.968799176] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518761.970406479] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518761.972143470] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518761.973764595] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518761.975488147] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518762.033390003] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518762.124546132] [ASR_Detect]: ----
[model_service-18] [INFO] [1788518762.215565532] [ASR_Detect]: ----
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